

A261 – Scale Hierarchy: From ξ to Masses, α , and E_0

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.1 Purpose

[STIPULATION] This document develops the complete scale hierarchy of FFGFT: from the single geometric constant ξ to the characteristic energy scales, to the lepton masses, to the fine-structure constant α , and to E_0 as the logarithmic mean. The core: all these quantities follow from ξ – but the connection is not always achievable by direct cancellation. Ratios (mass ratios) are correction-free; absolute values (including E_0 and α) need K_{frak} . *(Corrected on 30 Sept. 2026: previously mass ratios and E_0 were called correction-free – $E_0 = \sqrt{m_e m_\mu}$ is an absolute value; the corrected value is $E_0 / \sqrt{K_{\text{frak}}} = 7.398 \text{ MeV}$ (Doc. 190, R72).)*

.2 The Starting Point: A Single Constant

[B] FFGFT begins with a single dimensionless constant:

$$\xi = \frac{4}{3} \times 10^{-4} = 1.333 \dots \times 10^{-4}.$$

From it all fundamental scales follow in natural units ($\hbar = c = \ell_P = 1$).

[B] Characteristic T0 scales:

$$L_0 = \xi \cdot \ell_P \approx 1.333 \times 10^{-4} \ell_P, \quad E_0^{\text{scale}} = L_0^{-1} = \xi^{-1} \cdot E_P.$$

.3 Lepton Masses from Geometry

[K] The mass formulas (A100) in compact form:

$$m_e = c_e \cdot \xi^{5/2}, \quad m_\mu = c_\mu \cdot \xi^2, \quad m_\tau = c_\tau \cdot \xi^{3/2},$$

where the coefficients have geometric origin:

$$c_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}, \quad c_\mu = \frac{9}{4\pi\alpha}, \quad c_\tau = \frac{27\sqrt{3}}{8\pi\alpha^{3/2}}.$$

[K] Important: apparently simple fractions. In the short form the coefficients appear as rational numbers ($r_e = 4/3$, $r_\mu = 16/5$, $r_\tau = 25/9$ from A100). These fractions are *not* arbitrary and must not be directly cancelled: they encode spatial geometry (π , $\sqrt{3}$) and electromagnetic coupling (α). Both representations (Yukawa short form and extended form) are the same theory in two notations (A130). *(Note of 30 Sept. 2026: The extended form is not equivalent to A100. With $c_e = 9.681$, $c_\mu = 98.14$, $c_\tau = 2985$ one obtains $m_\mu/m_e = (c_\mu/c_e)\xi^{-1/2} = 878$ (measured 206.77) and $m_\tau/m_\mu = (c_\tau/c_\mu)\xi^{-1/2} = 2634$ (measured 16.82); the exponent difference τ/μ is $-1/2$ here instead of $-1/3$ as in A100 ($p = 3/2, 1, 2/3$). The mass ratios are not reproduced by this form; the ladder $m_i = r_i \xi^{p_i} v$ from A100 is authoritative.)*

.4 E_0 as Logarithmic Mean

[B] The characteristic energy E_0 is the geometric mean of electron and muon mass:

$$E_0 = \sqrt{m_e \cdot m_\mu}.$$

Logarithmically: $\log E_0 = \frac{1}{2}[\log m_e + \log m_\mu] - E_0$ lies exactly in the middle between m_e and m_μ on a logarithmic scale. Numerically: $E_0 = \sqrt{0.511 \times 105.658} \text{ MeV} \approx 7.35 \text{ MeV}$ (geometric mean). The fractally corrected value (A130, route 2) is $E_0 = 7.348/\sqrt{K_{\text{frak}}} \approx 7.398 \text{ MeV}$; this difference is where K_{frak} enters. *(Corrected on 30 Sept. 2026: previously "The value from the ξ -path (A130)" – that path, $E_0^2 = 4\sqrt{2} m_\mu/\xi^p$, has been removed from A130.)*

[B] From the mass formulas:

$$E_0 = \sqrt{c_e c_\mu} \cdot \xi^{9/4}.$$

E_0 is *not* correction-free: K_{frak} does not cancel in the geometric mean. The fractally corrected value is $E_0 = E_0^{\text{bare}}/\sqrt{K_{\text{frak}}} = 7.348/\sqrt{0.98667} = 7.397 \approx 7.398 \text{ MeV}$ (R72); the square root follows because E_0 enters α quadratically. *(Corrected on 30 Sept. 2026: previously correction-free and $E_0^{\text{exp}} = K_{\text{frak}} \cdot E_0^{\text{bare}} = 0.98667 \times 7.348 = 7.250 \text{ MeV}$ moves away from 7.398 MeV; the corpus relation is $E_0^{\text{bare}}/\sqrt{K_{\text{frak}}}$ (Doc. 190, R72).)*

.5 The Fine-Structure Constant from ξ

[K] The fundamental relation:

$$\alpha = \xi \cdot E_0^2 = c_e c_\mu \cdot \xi^{11/2}.$$

Substituting the geometric coefficients:

$$\alpha^{5/2} = \frac{27\sqrt{3}}{8\pi^2} \cdot \xi^{11/2}, \quad \text{hence} \quad \alpha = \left(\frac{27\sqrt{3}}{8\pi^2}\right)^{2/5} \cdot \xi^{11/5}.$$

With fractal correction (A040):

$$\alpha = \left(\frac{27\sqrt{3}}{8\pi^2}\right)^{2/5} \cdot \xi^{11/5} \cdot K_{\text{frak}}, \quad K_{\text{frak}} = 1 - 100\xi \approx 0.9867.$$

Numerically, however, this closed form with $\xi = \frac{4}{3} \times 10^{-4}$ gives $\alpha = 0.811 \times 2.98 \times 10^{-9} \times 0.9867 = 2.4 \times 10^{-9}$ ($\alpha^{-1} \approx 4.2 \times 10^8$), not $1/137$. The value $\alpha^{-1} = 137.035$ (0.0005 %) follows from $\alpha = \xi E_0^2$ with $E_0 = 7.398 \text{ MeV}$ (A130, R72). *(Corrected on 30 Sept. 2026: previously "Numerically: $\alpha^{-1} \approx 137.036$ – agreement to 0.0005 %" for the closed form – $(27\sqrt{3}/8\pi^2)^{2/5} = 0.811$, $\xi^{11/5} = 2.98 \times 10^{-9}$; the chain $\alpha = c_e c_\mu \xi^{11/2}$ does not reproduce α , cf. Doc. 070.)*

.6 Why α Does Not Follow Directly from $\xi^{11/2}$ Without Unit Convention

[S] $\alpha \propto \xi^{11/2}$ means: if ξ changed, α would change with exponent $11/2 = 5.5$. That is a strong coupling. The coefficients c_e, c_μ are of order 10^{19} in natural units – that is not a free parameter but the conversion factor between the Planck scale and MeV. In natural units (Planck energies) $c_e \sim 9.67$ and $c_\mu \sim 98.1$ – order 1 to 100, geometrically understandable. (Note of 30 Sept. 2026: $c_e \approx 9.68$ and $c_\mu \approx 98.1$ are the numbers of the coefficient formulas computed from α , not values in Planck units; in Planck units one would have $m_e/\xi^{5/2} = 4.19 \times 10^{-23}/2.05 \times 10^{-10} \approx 2.0 \times 10^{-13}$.)

.7 Ratios Are Correction-Free

[B] The mass ratio:

$$\frac{m_\mu}{m_e} = \frac{K_{\text{frak}} \cdot m_\mu^{\text{bare}}}{K_{\text{frak}} \cdot m_e^{\text{bare}}} = \frac{m_\mu^{\text{bare}}}{m_e^{\text{bare}}}.$$

K_{frak} cancels. The same holds for the Koide ratios (A110) and the g -2 ratios (A138). This is no coincidence: multiplicative corrections always cancel in ratios.

[K] Absolute values, by contrast, need K_{frak} : without correction $\alpha^{-1} \approx 138.9$ instead of $137.04 - 1.4\%$ deviation. With K_{frak} : $0.98667 \times 138.91 = 137.06 (+0.017\%)$. (Corrected on 30 Sept. 2026: previously “With K_{frak} : exact” – $K_{\text{frak}}/(\xi \cdot 7.3479^2) = 137.059$ against 137.036 .) Hence K_{frak} is not a post-correction but part of the structure (A040).

.8 The Complete Derivation Chain

[B] All fundamental quantities from ξ in one chain:

$$\begin{array}{c} \xi \xrightarrow{\text{geometry}} r_0, E_0^{\text{Scale}} \xrightarrow{\text{quantum numbers}} m_e, m_\mu, m_\tau \\ \xrightarrow{\text{geom. mean}} E_0 \xrightarrow{\xi \cdot E_0^2} \alpha \xrightarrow{\text{K-corr.}} \alpha_{\text{exp}} \end{array}$$

Additionally: $\xi \rightarrow L_0 \rightarrow G$ (gravitational constant, A142) and $\xi \rightarrow L_0$ (A015).

.9 Verification Script

- Numerically: $E_0 = \sqrt{m_e m_\mu}$, $\alpha = \xi E_0^2$, ratios correction-free vs. absolute values with K_{frak} :
python/A_Serie_Skripte/a261_skalenhierarchie.py

.10 Open Questions

The coefficients c_e, c_μ, c_τ are not fully independently derived. They contain α explicitly – and α in turn follows from the masses. The circularity is named as a methodological problem in A130; the T0 resolution (common geometric origin) is a claim, not a proof.

The quark coefficients are missing. The c_i structure for quarks is not worked out at the same level as for leptons (A150).

.11 Supersedes

This document subsumes: Doc. 070 (mathematical structure of T0 theory: circular ratios, scale hierarchy from ξ , lepton masses, α derivation, E_0 as logarithmic mean, geometric constant C , why ratios are correction-free). Doc. 093 (β -parameter derivation, self-consistency condition).

.12 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests with the author. Full wording of the rules in A010.